

Research article

Digital PCR-based assessment of pathogens in wastewater and antibiotic resistance genes in drinking water of the Kathmandu Valley, Nepal

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ABSTRACT

Digital polymerase chain reaction (dPCR) is a significant advancement in health-related water microbiology, enabling absolute quantification without standard curves. By partitioning samples into thousands of individual reactions, dPCR allows for precise quantification even in the presence of inhibitory substances common in environmental samples. This study evaluated the applicability of dPCR to detect gastroenteritis-causing enteropathogens (*Salmonella* spp., *Campylobacter coli*, *Campylobacter jejuni*, *Clostridium perfringens*, JC and BK polyomaviruses, and human adenovirus), crAssphage, and four antibiotic resistance genes (ARGs) (*sul1*, *bla_{NDM-1}*, *bla_{CTX-M}*, and *int11*) in wastewater and drinking water source samples. Wastewater samples were collected in 2018 from two municipal wastewater treatment plants (WWTPs) that used an oxidation ditch system ($n = 12$) and a stabilization pond system ($n = 10$) in the Kathmandu Valley, Nepal, whereas the drinking water source samples were collected from shallow wells between 2015 and 2016 ($n = 22$). The enteropathogens and ARGs were analyzed using the QIAcuity dPCR System. The highest detection ratio was observed by crAssphage in wastewater (100 %, 22/22) and by *sul1* and *int11* in drinking water sources (91 %, 20/22). The log₁₀ reduction values evaluated in both WWTPs were <1 using dPCR, consistent with those obtained in previous quantitative PCR studies, and may be used to cross-validate between methods. However, this study also observed low detection ratios and concentrations of enteropathogens, likely due to factors such as low sample volumes, dead volume, thermal conditions, and the dPCR platform used. Thus, optimizing these variables is imperative to enhance the applicability of dPCR in environmental assessments.

1. Introduction

Pathogen detection is imperative in microbiological research, as it significantly impacts human and environmental health (Lazcka et al., 2007). Culture-based methods, immunology-based methods, and quantitative polymerase chain reaction (qPCR) are commonly used to detect pathogens in human and environmental samples (Amani et al., 2015). These methods are selected due to their sensitivity, specificity, accuracy, cost-effectiveness, rapidity, and ability to detect minute amounts of pathogenic markers regardless of pathogen viability (Amani et al., 2015). Among these methods, qPCR is currently the preferred method for identifying microorganisms in environmental samples due to its advantages and flexibility of the primer/probe design, which allows for

targeting a wider range or more specific viruses (Colinon et al., 2013). However, there are challenges to analyzing environmental waters, as inhibitory substances, such as humic acid and fulvic acid, can coexist with pathogens and interfere with qPCR (Haramoto et al., 2018; Matheson et al., 2010). This necessitates further purification or dilution of the nucleic acid before performing qPCR, but these additional steps can lead to further loss of nucleic acids, which may not be possible for samples with low levels of the target (Haramoto et al., 2018; Matheson et al., 2010). Furthermore, the evaluation of the target concentration depends on standard curves produced during a qPCR run, limiting the comparability of results among various qPCR thermocyclers (Malla et al., 2024; Pecson et al., 2021).

Digital PCR (dPCR) is a promising alternative to qPCR to combat

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these challenges. This method partitions the sample into thousands of individual reactions, allowing simultaneous amplification and detection of target molecules (Cavé et al., 2016; Salipante and Jerome, 2020). The core principle of dPCR involves partitioning DNA or RNA samples into thousands to millions of independent micro-reactions, such that each partition ideally contains one or zero copies of the target (Lei et al., 2021; Tan et al., 2021). Once partitioned, PCR amplification parallelly occurs across all chambers. After thermal cycling, each partition is analyzed for the presence (positive) or absence (negative) of fluorescence, which indicates whether amplification of the target sequence occurred (Kopylova et al., 2023). Unlike qPCR, which relies on the exponential phase of amplification and comparison to reference standards, dPCR performs endpoint detection and binary classification of each reaction. The number of positive reactions is then used to calculate the absolute concentration of the target based on Poisson distribution statistics, which accounts for the probability of multiple targets occupying a single partition (Mirabile et al., 2024; Quan et al., 2018; Tan et al., 2021). This method enhances accuracy, especially for detecting low-abundance targets or subtle fold changes and is particularly robust in the presence of PCR inhibitors, which are common in environmental samples (Capo et al., 2021). This high-throughput approach offers several advantages to detect and quantify waterborne pathogens and antibiotic resistance genes (ARGs), including increased sensitivity to inhibitory substances, accuracy, and reproducibility (Capo et al., 2021; Tiwari et al., 2022). Given its versatility, dPCR has been used in various applications, including cancer studies, fish population determination in humic-rich ecosystems, pathogen detection in root and soil samples, microorganism analysis in aquatic environments, clinical microbiology, avian virology, food industry, oncology, and health-related water microbiology (Ahmed et al., 2022a, 2022b; Boogaerts et al., 2021; Capo et al., 2021; Ho et al., 2022; Isham et al., 2024; Kokkoris et al., 2021; Li et al., 2022; Mao et al., 2019; Tiwari et al., 2022; Tong et al., 2017).

A persistent public health issue in several developing countries has been acute gastroenteritis (AGE) (Chitambar et al., 2012; Hossian et al., 2021; Sthapit et al., 2024; Tandukar et al., 2019; Udompat et al., 2024). With the increase in antibiotic prescriptions, ARGs have also developed into an emerging global public health issue (Gwenzi et al., 2020; Siri et al., 2024; Thakali et al., 2021; Zhang et al., 2021, 2022). Recent studies have implemented wastewater-based epidemiology for the surveillance of pathogens that induce outbreaks (Haramoto et al., 2020; Inson et al., 2024; Randazzo et al., 2020; Tandukar et al., 2022). The high sensitivity and tolerance to inhibitors of dPCR in wastewater and environmental samples makes it ideal for detecting and monitoring gastroenteritis-causing pathogens and ARGs, allowing for rapid public health responses to prevent their spread and mitigate these issues. Assessment of these targets, particularly gastroenteritis-causing pathogens, has been performed only using qPCR, enzyme-linked immunosorbent assay, or culture-based methods in the Kathmandu Valley, Nepal (Ansari et al., 2013; Cardemil et al., 2017; Sherchand et al., 2012; Shrestha et al., 2018, 2019; Sthapit et al., 2024; Tandukar et al., 2021). Only one other study has analyzed the drinking water sources for ARGs in the Kathmandu Valley using dPCR (Amarasiri et al., 2022). Given that numerous studies globally have determined dPCR to be useful for wastewater-based epidemiology, this study aimed to assess the applicability of dPCR for analyzing wastewater and drinking water source samples from shallow wells in the Kathmandu Valley.

2. Materials and methods

2.1. Sample collection

This study assessed 22 wastewater samples collected by grab sampling during the morning hours (9:00 to 11:00 a.m.) in 2018 from two wastewater treatment plants (WWTPs) in the Kathmandu Valley: WWTP A (n = 12) and WWTP B (n = 10). For each plant, both influent and effluent samples were collected, comprising six influent and six effluent

samples from WWTP A, and five influent and five effluent samples from WWTP B, as described previously (Sthapit et al., 2022, 2024; Tandukar et al., 2021). Influent samples refer to the raw wastewater entering the WWTPs, while effluent samples refer to the treated water prior to discharge into nearby rivers. WWTP A, designed for a population of 74,000 with a capacity of 17.3 million liters daily (MLD), operates using an oxidation ditch system. In this plant, the influent undergoes screening, grit removal, and biological treatment in an oxidation ditch, followed by secondary clarification and flow measurement before effluent discharge. In contrast, WWTP B, serving the eastern Lalitpur population with a capacity of 1.1 MLD, employs a series of stabilization ponds. In WWTP B, the influent flows through anaerobic, facultative, and maturation ponds, where treatment occurs through microbial degradation, sedimentation, and sunlight-driven pathogen removal prior to effluent discharge. This study analyzed drinking water source samples (shallow well samples) collected in the Kathmandu Valley between 2015 and 2016 (n = 22), as described previously (Malla et al., 2018a; Sthapit et al., 2020; Thakali et al., 2022). Wastewater samples were collected in sterile 1 L plastic bottles, and groundwater samples were collected in sterile 100 mL plastic bottles and transported to the laboratory in an icebox within 6 h of collection for further processing.

2.2. Bacterial/viral concentration and DNA extraction

As reported previously (Ghaju Shrestha et al., 2017; Malla et al., 2018a, 2018b), 10 mL wastewater samples and 100 mL drinking water source samples were passed through a membrane filter (pore size, 0.22 µm; diameter, 47 mm; Nalgene, Tokyo, Japan) and processed to acquire the bacterial concentrate, which was subjected to the CicaGeneus DNA Extraction Kit (Kanto Chemical, Tokyo, Japan) to obtain 200 µL DNA extract. In contrast, 50 mL wastewater samples were concentrated using the electronegative membrane-vortex method (Haramoto et al., 2012) to obtain the viral concentrate, which underwent DNA extraction using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) to obtain 200 µL DNA extract.

2.3. Digital PCR

This study analyzed the detection of four pathogenic bacteria [*Salmonella* spp. (*invA*) (Iijima et al., 2004), *Campylobacter jejuni* (*gyrA*) (Iijima et al., 2004), *Campylobacter coli* (*ceuE*) (Best et al., 2003), and *Clostridium perfringens* (*cpe*) (Albini et al., 2008)] and four viruses [JC polyomavirus (JCPyV) (Pal et al., 2006), BK polyomavirus (BKPv) (Pal et al., 2006), human adenovirus (HuAdV) (Heim et al., 2003), and crAssphage (Stachler et al., 2017)] in wastewater samples, as shown in Fig. 1. In drinking water source samples, the analysis focused on key ARGs, including the sulfonamide-resistant gene (*sul1*) (Czekalski et al., 2012), the New Delhi metallo-β-lactamase-1 gene (*bla_{NDM-1}*) (Naas et al., 2011), and the CTX-M-β-lactamase gene (*bla_{CTX-M}*) (Colomer-Lluch et al., 2011), as well as on the class 1 integrase genes (*int1*) (Barraud et al., 2010), which is highly persistent in the environment and proposed to be an ARG indicator (Gillings et al., 2015), thereby considered an ARG assay in this study. Initially, the dPCR performance of the assays and markers was evaluated using five 10-fold serial dilutions of the standards from 1.0×10^0 to 1.0×10^5 copies/µL, which are commonly used in qPCR, using the QIAcuity dPCR System (Qiagen). For bacterial DNA, 3.6 µL (30 %) was added as a sample in each well, whereas 6.0 µL (50 %) was added for viral DNA. This method analyzed wastewater samples in singlicate in the Nanoplate 8.5K 96-well (Qiagen), which consists of 96 wells, each with 8500 partitions or simultaneous reactions. The preparation involved adding 8.4 µL/well reaction mixture for pathogenic bacteria and 6.0 µL/well reaction mixture for DNA viruses with an additional 20 % volume for both. The 8.4 µL reaction mixture comprised 3.0 µL of 4 × Probe PCR Master Mix (Qiagen), 0.1 µL each of forward and reverse primers (100 µM), 0.05 µL probe (100 µM), and 5.15 µL RNase-free water to which 3.6 µL DNA would be transferred,

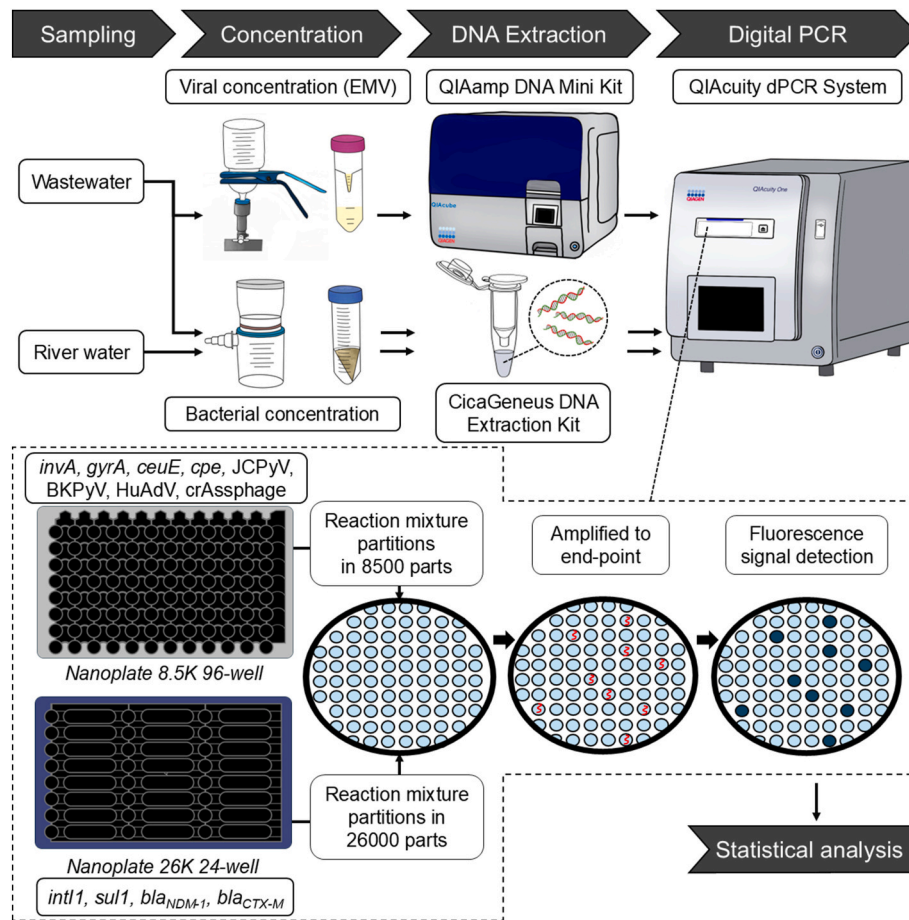


Fig. 1. Workflow of dPCR-based detection of pathogenic bacteria and viruses from wastewater and ARGs from drinking water samples.

whereas the 6.0 μL reaction mixture comprised 3.0 μL of $4 \times$ Probe PCR Master Mix, 0.1 μL each of forward and reverse primers (100 μM), 0.05 μL probe (100 μM), and 2.75 μL RNase-free water to which 6.0 μL DNA would be added resulting in a final volume of 12.0 μL /well mixture. The final mixture was transferred to the Nanoplate, ensuring that no bubbles were formed in the wells and finally the Nanoplate was sealed. This approach follows validated methodologies for digital PCR optimization, including reaction volume setup, valid partition thresholds, and manual threshold adjustments, as recommended in previous studies using the QIAcuity dPCR system (Malla et al., 2024).

On the other hand, drinking water source samples were assessed in singlicate using a Nanoplate 26K 24-well (Qiagen), which consists of 24 wells, each with 26,000 partitions or simultaneous reactions. In this plate, a 44.0 μL pre-reaction mixture comprising 13.2 μL DNA, 11.0 μL of $4 \times$ Probe PCR Master Mix, 0.352 μL each of forward and reverse primer (100 μM), 0.176 μL probe (100 μM), and 18.92 μL RNase-free water was prepared. Then, the pipette was set at 42.0 μL , but transferred 40 μL of the prepared PCR mixture with reverse pipetting in the Nanoplate, which was sealed and subjected to dPCR. The thermal conditions for wastewater and drinking water source samples were as follows: 95 $^{\circ}\text{C}$ for 2 min, followed by 40 cycles of 95 $^{\circ}\text{C}$ for 5 s and 60 $^{\circ}\text{C}$ for 30 s. Negative and positive controls comprised RNase-free water and the gBlocks Gene Fragments (Integrated DNA Technologies, Coralville, LA, USA) or the artificially synthesized plasmid DNA (Eurofins Genomics, Tokyo, Japan), respectively.

For the analysis, only wells with a minimum of 20 % negative partitions were considered. Furthermore, as suggested by the guidelines for the minimum information for publication of quantitative digital PCR experiments (Huggett, 2020), the number of valid partitions for the Nanoplate 8.5K 96-well ranged from 7751 to 8298 (91 %–98 %) and for

the Nanoplate 26K 24-well ranged from 24,558 to 25,487 (94 %–96 %), for which the reaction volumes were 0.34 nL and 0.82 nL, respectively. Thus, the dead volumes calculated in this study were 2.6–2.8 μL (22 %–24 %) for the Nanoplate 8.5K 96-well and 20.1–20.9 μL (50 %–52 %) for the Nanoplate 26K 24-well. The thresholds for all wells were adjusted manually.

2.4. Statistical analysis

Fisher's exact test was performed to evaluate the difference in positive ratios between influent and effluent, the paired *t*-test was performed to determine the differences in mean concentrations between influent and effluent, and the Welch *t*-test was used to compare the concentrations between WWTPs. The $\log_{10}$ reduction values (LRVs) were obtained by evaluating the difference in logarithmic concentrations between the influent and effluent when at least the influent was positive using the limit of detection (LOD) values for non-detect samples. Samples with more than three positive partitions were classified as positive, while those with three or fewer were deemed negative. The LOD was established at 10 % of the measured value. To evaluate the correlation between pathogenic bacterial and viral assays, Pearson's correlation was performed. Absolute quantification by dPCR was performed using the Poisson distribution, which estimates the most probable number (MPN) of target copies based on the proportion of positive partitions (Tiwaree et al., 2022). $p < 0.05$ was considered statistically significant. Statistical analyses were performed using Microsoft Excel for Microsoft 365 (Microsoft Corporation, Redmond, WA, USA), and the figures used in this study were visualized using R version 4.3.3 (The R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Performance of bacterial, viral, and ARG assays using dPCR

The performance of bacterial, viral, and ARG assays was evaluated by testing the positive controls on dPCR for linearity of amplification. Pathogenic bacteria, BKPyV, and ARG assays, except for *bla*_{NDM-1}, detected four of five dilutions of standard samples, whereas viruses, except for BKPyV and *bla*_{NDM-1}, were detected in only three of five dilutions. In Fig. 2, the performance of these assays covered a dynamic range corresponding to up to four dilution factors, demonstrating their capability to accurately quantify targets across a broad range of concentrations. A coefficient of determination (R^2) value from 0.997 to 1.000 was exhibited by pathogenic bacterial assays, from 0.988 to 1.000 was observed in viral assays, and from 0.999 to 1.000 was observed in ARG assays using dPCR. The consistent linear performance of all targets indicated the reliability of dPCR; thus, dPCR was employed to assess wastewater and drinking water source samples for pathogenic bacteria and viruses and ARGs, respectively.

3.2. Detection of pathogenic bacteria and viruses in WWTPs using dPCR

In Table 1, all pathogenic bacteria were detected in influents of both WWTPs using dPCR. Only *invA* and *ceuE* were detected in the WWTP A effluent, whereas all targets, except for *invA*, were detected in the WWTP B effluent. Among these targets, *cpe* in WWTP A exhibited a significant difference in the detection ratio between influent and effluent ($p < 0.05$, Fisher's exact test), whereas others showed no significant difference. Fig. 3 presents the mean pathogenic bacterial concentrations in influents and effluents in both WWTPs. Among the four assays, *gyrA*, *ceuE*, and *cpe* exhibited significantly higher mean concentrations in the WWTP A influent compared to effluent, whereas only *invA* observed a significant

difference between influent and effluent in WWTP B ($p < 0.05$, paired t -test). Between the WWTPs, *gyrA* and *cpe* exhibited a significantly lower mean concentration in the WWTP A effluent compared to WWTP B ($p < 0.05$, Welch t -test), whereas no significant differences were observed in influents between the WWTPs. Table 2 shows the LRVs evaluated in the two WWTPs, where a maximum LRV of 1.44 and a minimum of -0.10 was observed for pathogenic bacteria in WWTPs.

In addition to pathogenic bacteria, three viral assays were tested in wastewater samples using dPCR. As shown in Table 1, all four viruses were detected in at least 33 % of the influents and effluents from both WWTPs, with crAssphage detected in all samples. Of the four viruses, only JCPyV observed a significantly higher mean concentration in influents of WWTP A compared to the effluents ($p < 0.05$, paired t -test), whereas no significant difference was exhibited between the influent and effluent of WWTP B. Compared to pathogenic bacteria, only one virus, JCPyV, showed a significant difference in mean concentration between the effluents of the two WWTPs ($p < 0.05$, Welch t -test). In Table 2, the maximum LRV of viruses observed in WWTPs was 1.34, and the minimum was -0.35 for viruses. Although the concentrations of the pathogenic bacteria and viruses appeared to vary across sampling dates, the limited number of sampling dates (six in 2018) was insufficient to perform time-series analysis. More frequent sampling would be required to reliably capture temporal trends in the Kathmandu Valley.

3.3. Detection of ARGs and *int1* in drinking water source samples using dPCR

Subsequently, three ARGs and *int1* in shallow well samples were detected using dPCR. As shown in Table 3, all, except for *bla*_{NDM-1}, were detected in drinking water source samples. *sul1* and *int1* were detected in 91 % (20/22) of the samples with mean concentrations of $\sim 7 \log_{10}$ copies/100 mL. The *bla*_{CTX-M} assay observed the lowest concentration

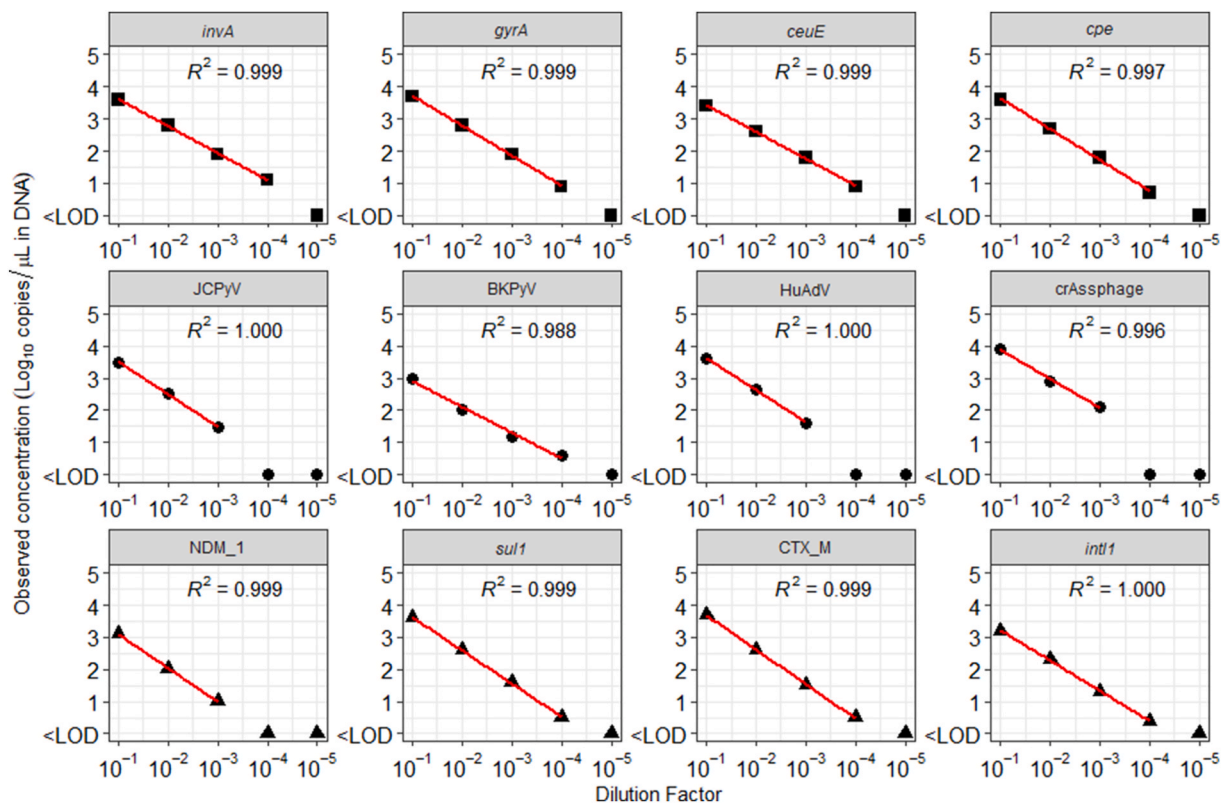


Fig. 2. Regression plots showing the relationship between dilutions of standard samples and measured concentrations for pathogenic bacteria (squares), viruses (circles), and ARG (triangles) assays. The x-axis represents the serial dilutions, while the y-axis displays the corresponding concentrations, illustrating the linearity and sensitivity of each assay.

Table 1

Detection ratios of pathogenic bacteria and viruses in influents and effluents of WWTPs A and B.

WWTP	Sample type	No. tested samples	No. positive samples (%)							
			Pathogenic bacteria				Virus			
			<i>invA</i>	<i>gyrA</i>	<i>ceuE</i>	<i>cpe</i>	JCPyV	BKPyV	HuAdV	crAssphage
A	Influent	6	2 (33)	4 (67)	4 (67)	5 (83)	6 (100)	5 (83)	5 (83)	6 (100)
	Effluent	6	5 (83)	0 (0)	1 (17)	0 (0)	3 (50)	2 (33)	5 (83)	6 (100)
B	Influent	5	4 (80)	5 (100)	2 (40)	4 (80)	4 (80)	4 (80)	4 (80)	5 (100)
	Effluent	5	0 (0)	4 (80)	2 (40)	4 (80)	5 (100)	4 (80)	5 (100)	5 (100)
Total		22	11 (50)	13 (59)	9 (41)	13 (59)	18 (82)	15 (68)	19 (86)	22 (100)

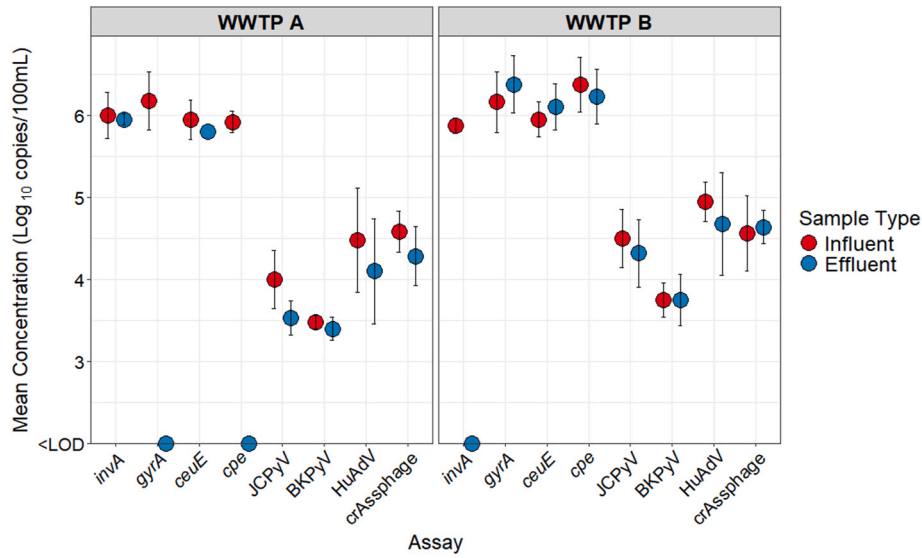


Fig. 3. Pathogenic bacterial and viral concentrations in influents and effluents of WWTPs A and B.

Table 2

LRVs of pathogenic bacteria and viruses in WWTPs A and B.

WWTP	LRV (mean $\pm$ standard deviation) (No. of sample pairs)							
	Pathogenic bacteria				Virus			
	<i>invA</i>	<i>gyrA</i>	<i>ceuE</i>	<i>cpe</i>	JCPyV	BKPyV	HuAdV	crAssphage
A	0.66 $\pm$ 0.52 (2)	1.44 $\pm$ 0.32 (4)	0.94 $\pm$ 0.28 (4)	1.18 $\pm$ 0.15 (5)	1.34 $\pm$ 1.23 (6)	1.04 $\pm$ 1.00 (5)	0.34 $\pm$ 1.42 (6)	0.32 $\pm$ 0.44 (6)
B	1.13 $\pm$ 0.12 (4)	0.12 $\pm$ 0.65 (5)	-0.10 $\pm$ 0.04 (2)	0.45 $\pm$ 0.52 (4)	-0.35 $\pm$ 0.92 (5)	0.00 $\pm$ 1.37 (5)	-0.32 $\pm$ 1.40 (5)	-0.07 $\pm$ 0.33 (5)

Table 3

Positive ratios and mean concentrations of ARGs and *intI1* in shallow well samples using dPCR.

Target	No. of tested samples	No. of positive samples (%)	Concentration (mean $\pm$ standard deviation) ($\log_{10}$ copies/100 mL)
<i>bla</i> _{NDM-1}	22	0 (0)	Not applicable
<i>sul1</i>	22	20 (91)	7.5 $\pm$ 1.1
<i>bla</i> _{CTX-M}	22	1 (5)	4.4
<i>intI1</i>	22	20 (91)	7.3 $\pm$ 1.1

(4.4 $\log_{10}$ copies/100 mL). Total coliforms and *E. coli* were measured in the groundwater samples, with mean concentrations of 4.4 $\pm$ 1.6 $\log_{10}$ MPN/100 mL and 2.6 $\pm$ 1.6 $\log_{10}$ MPN/100 mL, respectively, as previously reported (Thakali et al., 2022). The presence of these fecal indicators supports the potential for microbial contamination in the groundwater, which corresponds to the high detection frequencies of *intI1* and *sul1*, indicating high contamination from sulfonamide usage.

These findings may also imply presence of other ARGs that were not targeted in this study, although resistance to last-resort antibiotics, such as carbapenems or extended-spectrum β -lactamases, was not observed. Additionally, although the groundwater samples were collected on different dates, they originated from various locations across the Kathmandu Valley, making it unsuitable to conduct reliable time-series analysis of ARG concentrations.

3.4. Correlation analysis of pathogenic bacteria and viruses

Pearson's correlation analysis was conducted to evaluate the relationships between the pathogenic bacteria and viruses. Strong positive correlations ($r \geq 0.70$, $p < 0.05$) were observed in five pairs, whereas moderate positive correlations ($0.40 \leq r < 0.70$, $p < 0.05$) were found in eight pairs (Fig. 4). The strongest correlation was between *cpe* and *gyrA* ($r = 0.87$, $p < 0.05$). Notably, JCPyV observed the greatest number of correlations, showing significant positive associations with *gyrA* ($r = 0.62$), *cpe* ($r = 0.61$), BKPyV ($r = 0.72$), HuAdV ($r = 0.78$), and crAssphage ($r = 0.79$).

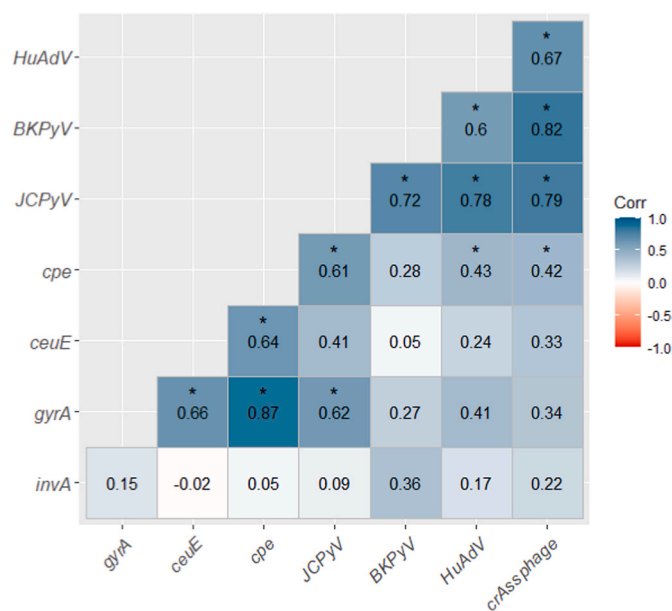


Fig. 4. Pearson's correlation analysis between pathogenic bacteria and viruses in wastewater. Boxes with asterisks indicate significance ($p < 0.05$).

4. Discussion

Conventionally, qPCR has been recognized as the most effective tool for pathogen detection in environmental samples due to its efficiency compared to other rapid methods (Monteiro and Santos, 2017). However, qPCR results can be influenced by inhibitory substances in environmental samples, whereas dPCR is generally more tolerant to such inhibitors (Rački et al., 2014). Furthermore, the accuracy of quantitative data in qPCR is inherently dependent on the precision of the standards, whereas dPCR offers absolute quantification, mitigating reliance on a standard curve (Maheshwari et al., 2017). Therefore, this study assessed the effectiveness of dPCR to detect pathogens and ARGs in wastewater and river water samples, respectively, in the Kathmandu Valley.

The performance of standard samples of target assays was initially evaluated to determine their suitability for dPCR application. The dynamic ranges of these assays demonstrated a maximum of three orders of magnitude for *sul1* and *bla*_{CTX-M} assays. Validation of dynamic range is essential for any method to ensure alignment with the intended use of the assay (Hays et al., 2024). However, the relatively low dynamic range observed with dPCR in this study represents a limitation. Despite this, pathogenic bacterial, viral, and ARG assays demonstrated good linearity of amplification, indicating reliable and consistent performance when using dPCR. A similar study evaluating the *invA* assay with dPCR reported congruent R^2 values in this study (Jikumaru et al., 2020). The parameters measured in this study adhered to criteria established by several relevant publications and guidelines on PCR experiments (Bustin et al., 2009; Hays et al., 2022; Ma et al., 2021), suggesting the feasibility of dPCR for use in environmental sample analysis.

The pathogenic bacteria and viruses investigated in this study are common etiological agents of gastroenteritis, making their expected presence in wastewater. Wastewater samples in this study were previously tested for respective targets using qPCR (Sthapit et al., 2024; Tandukar et al., 2021). Both studies found significant differences in the detection ratio for *cpe* in WWTP A, whereas other assays showed no significant differences using dPCR. However, pathogenic bacterial assays demonstrating a significant difference in concentration between influent and effluent in this study were analogous to those in a previous study (Sthapit et al., 2024). This consistency suggested that qPCR and dPCR are consistent in their ability to detect and quantify the targets and that results from one method validate the findings of the other,

enhancing the overall data reliability.

The individual results of pathogenic bacterial and viral assays using qPCR exhibited higher positive ratios and concentrations than those obtained using dPCR, a trend consistent with previous studies (D'Aoust et al., 2021; Hinkle et al., 2022; Steele et al., 2021). One possible factor influencing these results is the difference in total reaction and sample volumes between the methods. A total reaction volume of 25 μ L with a sample volume of 2.5 μ L was used in qPCR. In contrast, dPCR employed a reaction volume of 12 μ L for pathogenic bacteria and viruses, with sample volumes of 3.6 μ L for pathogenic bacteria and 6.0 μ L for viruses, and 40 μ L for ARGs, where the sample volume was 12 μ L. Despite the higher sample-to-reaction volume in dPCR, the assays exhibited lower sensitivity, as indicated by lower detection rates and concentrations (Acosta Soto et al., 2017; Hayden et al., 2013). Nevertheless, in dPCR, the partitioning of samples into partitions or droplets <1 nL acts as independent reactions, reducing the overall impact of inhibitors on amplification and ensuring reliable quantification of targets. In contrast, qPCR does not involve partitioning, making the quantification of targets more susceptible to suppression in amplification by PCR inhibitors compared to dPCR. Another factor contributing to the difference in results is the potential overestimation of concentrations in qPCR, which may produce an inflated standard curve. This overestimation could arise from DNA adhering to the plastic tube walls or degraded DNA samples, leading to overquantification of targets in wastewater by qPCR and potential underquantification by dPCR (Hinkle et al., 2022). Additionally, dead volume, which refers to the reaction volume lost between sample preparation and final fluorescence reading, may also affect sensitivity by not accounting for the template in the lost volume. This dead volume is particularly significant in dPCR, with QIAcuity dPCR showing a 24 % loss using a Nanoplate 8.5K 96-well and a 52 % loss using a Nanoplate 26K 24-well. Finally, the thermal conditions set for dPCR were identical for all assays, whereas they were optimized for each target in qPCR. This difference in thermal optimization could have further influenced the results.

WWTPs from which wastewater samples were collected lack a disinfection step and are inadequate in treating influent, as evidenced by multiple studies (Sthapit et al., 2022, 2024; Tandukar et al., 2021; Thakali et al., 2021). This study supported this finding, where LRVs for pathogenic bacteria and viruses using dPCR mirrored those obtained from previous studies (Tandukar et al., 2021; Sthapit et al., 2024). LRVs of ~ 1 were observed for all targets using qPCR and dPCR, indicating comparable sensitivities and similar detection efficiencies between the methods. Similar LRVs also serve as cross-validation, suggesting that the quantification results obtained from one method align with those from the other, highlighting the ineffectiveness of WWTPs in reducing pathogens in the influent. One study suggested that existing treatment technologies fail to remove ARGs, contributing to their spread in the environment through effluents and contaminated river water (Thakali et al., 2021), which inadvertently affects groundwater. In this study, the co-detection of *int1* and *sul1* in groundwater indicates potential anthropogenic pollution and may also reflect in-system proliferation of ARGs (Haenelt et al., 2023; Lee et al., 2021). While this study did not assess a wider range of ARGs or additional environmental parameters, the detection of *int1* supports that other ARGs may also be present in the groundwater in the Kathmandu Valley (Wang et al., 2025). Future studies should include a broader list of ARGs and various sample types to better characterize resistance dissemination pathways. Groundwater samples in this study were also previously tested for corresponding targets using qPCR (Thakali et al., 2022). As determined by qPCR, ARG detection and concentrations in drinking water source samples were higher than those observed with dPCR. This discrepancy may be attributed to factors such as sample volume, dead volume, thermal conditions, and potential overestimation of concentrations by qPCR, as mentioned above. A similar study using river water and groundwater samples from the Kathmandu Valley reported ~ 6 log₁₀ copies/100 mL concentrations for *sul1* and *int1* in shallow dug well samples (Thakali

et al., 2021). In contrast, this study exhibited concentrations $\sim 1 \log_{10}$ higher than those reported by the other study, which utilized the X200 AutoDG droplet dPCR system (Bio-Rad, Hercules, CA, USA) (Amarasiri et al., 2022). This difference may be due to dPCR platform variations, including partitioning methods, quantification accuracy, instrument design, and detection chemistries. Moreover, differences in reagents and their composition and ability to mitigate inhibition can vary between dPCR platforms. For instance, a study in Ottawa and Quebec, Canada, observed higher target concentrations using qPCR than dPCR, even after two- and fivefold dilutions. The study suggested that variations in reagents and their effectiveness in overcoming inhibition could influence the observed results, particularly when target concentrations are close to LOD (D'Aoust et al., 2021). To further explore the applicability of dPCR, comparisons with other commonly used molecular methods have been explored. A comparative study on the detection of human papilloma-virus (HPV) using next-generation sequencing (NGS), droplet dPCR, and qPCR demonstrated that both NGS and droplet dPCR exhibited comparable sensitivity for detecting HPV DNA in clinical samples. The authors concluded that dPCR is well-suited for accurate quantification of target copy numbers, qPCR is preferable when targets are present in high abundance, and NGS is most appropriate when both target detection and mutation profiling are required (Mattox et al., 2022). Similarly, Maestre-Carballe et al. (2024) conducted a side-by-side comparison of ARG abundance using metagenomic sequencing and dPCR, reporting comparable trends in ARG profiles and detection across samples. However, dPCR showed higher sensitivity and accuracy in samples with low ARG abundance, whereas metagenomics offered a broader view of ARG diversity. These findings highlighted both the strengths and limitations of dPCR and other culture-independent methods, such as dPCR's dependence on primer specificity and the need to meet bioinformatics thresholds for accurate ARG identification in metagenomic analysis.

In addition to examining the occurrence of pathogens and ARGs using dPCR, this study assessed the correlations between pathogenic bacteria and viruses to determine their interrelationships. Among all targets, JCPyV exhibited the highest number of significant correlations, including two pathogenic bacteria and all potential indicator viruses. Similarly, crAssphage demonstrated strong correlations with all viruses. For a virus to be considered a reliable fecal indicator, it must (i) be consistently detected in the influent sample, (ii) be at least as resistant to treatment processes as pathogenic viruses being monitored, and (iii) remain detectable in the treated effluent in WWTPs (Sabar et al., 2022). In this study, crAssphage fulfilled all these criteria, with 100 % detection across all samples (22/22), high concentration levels in both influent and effluent, and low LRVs compared to the tested pathogenic viruses. Furthermore, previous studies revealed that crAssphage abundance is not significantly affected by population coverage or WWTP capacity (Crank et al., 2020), and exhibits minimal seasonal variation (Ahmed et al., 2021). These findings suggest that crAssphage is a robust and conservative marker of fecal contamination and viral reduction performance. These results are consistent with previous qPCR-based studies conducted in New Orleans (Tandukar et al., 2020), North Wales (Farkas et al., 2019), and Italy (Crank et al., 2020), further validating the applicability of crAssphage as an indicator for tracking human fecal pollution.

Although this study demonstrated the potential of dPCR as a valuable tool to detect pathogens and ARGs in environmental samples, it also highlighted the importance of considering experimental factors, such as sample volume, reaction conditions, and platform-specific differences. Despite the promising results, the observed discrepancies between qPCR and dPCR underscore the need for further optimization and standardization of dPCR protocols to ensure consistent and reliable data. Future research should focus on refining these methodologies, particularly in addressing the challenges posed by PCR inhibitors and enhancing sensitivity in complex environmental matrices. Exploring the comparative performance of different dPCR platforms across various environmental contexts is crucial in establishing dPCR as a robust and versatile

tool for environmental monitoring and public health assessment. By addressing these challenges, dPCR could play a pivotal role in advancing the understanding of microbial dynamics in environmental systems and improving water quality and safety management.

5. Conclusion

In recent years, dPCR has gained popularity due to its robustness against inhibitory factors in environmental samples and its capability for absolute quantification. This study assessed the performance of dPCR in detecting pathogenic bacteria, viruses, and ARGs and subsequently applied it to wastewater and drinking water source samples. Results demonstrated good linearity for all targets, indicating dPCR reliability. Among targets, crAssphage exhibited the highest detection ratio in wastewater and strong correlations with other viruses, consistent with previous studies, underscoring its potential as a fecal indicator. However, LRVs for pathogenic bacteria and viruses in both WWTPs were <1 , comparable to results from qPCR studies. This suggested comparable sensitivities and detection efficiencies between the methods but highlighted some limitations of dPCR such as lower detection ratios for certain targets compared to qPCR. The discrepancies may be attributed to dynamic range limitations, dead volume, sample volume, thermal conditions, and partitioning. Future research should focus on standardizing dPCR protocols and improving reagent formulations to reduce inhibition effects, thereby enhancing its robustness for environmental monitoring and public health applications.

CRedit authorship contribution statement

Niva Sthapit: Writing – original draft, Investigation, Formal analysis. **Ocean Thakali:** Writing – review & editing, Investigation, Formal analysis. **Bikash Malla:** Writing – review & editing, Investigation. **Sunayana Raya:** Investigation. **Jeevan B. Sherchand:** Resources. **Eiji Haramoto:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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